

Data Science

Autoregressive models

Estimating linear temporal latent factors

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Table of contents

Motivation

Temporal series

Trend, seasonality and residual

Prediction

Stationary process

Exponential smoothing

AR models

Order selection

Linear Predictive Coding (LPC)

What is the lecture about?

- ★ To understand **temporal data** modeling
- ★ To understand temporal data analysis
- ★ To understand the concept of **auto**-regression
- linear temporal latent model
- ★ To present the basic of autoregressive models ($\text{AR}(p)$)

Reading: [3] and [1] (chap 13)

Definition

- ★ A temporal series is a sequence of observations depending on time

$$\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_t, \dots, \mathbf{y}_T \quad \mathbf{y}_t \in \Omega \subseteq \mathbb{R}^D$$

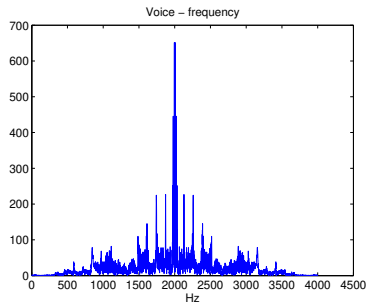
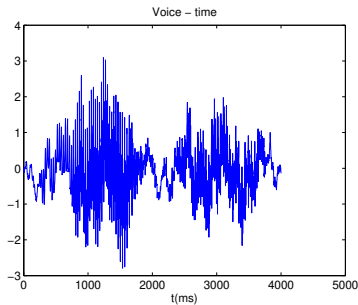
- ★ In general, we assume a constant time interval
 - ★ For statistical temporal data analysis, we study **causality**
- ⇒ $\mathbf{y}_t$ depends on the p preceding values $\mathbf{y}_{t-1}, \dots, \mathbf{y}_{t-p}$
- ★ Causality is the information we want to model (give/measures parameters)

Applications

- ★ Économetry (trend prediction, risk analysis...)
- ★ Social (demography, migrations ...)
- ★ Physical measures (explanation of physical constants/values)
- ★ Communication and information → telecom, network, coding, speech, video ...

Examples

- ★ Temporal data analysis (mean, variance, correlation)
- ★ Frequency analysis (periodicity)
- ★ Combination time/frequency, time/scale
- Non-stationary series



Temporal series

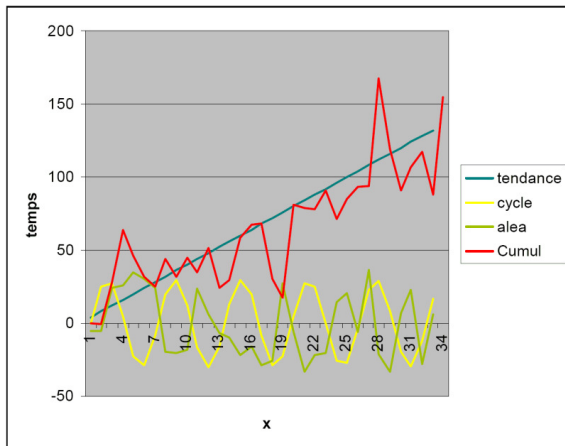
★ Definition

$$\mathbf{y}_t = \mathbf{g}(t) + \epsilon_t, \quad t \in \llbracket T \rrbracket$$

- ★ $\mathbf{y}_t$ scalar or vector
 - ★ $\mathbf{g}(t)$ **structure** of the series
 - ★ ϵ_t noise **centered, uncorrelated** (white noise)
- ⇒ ϵ_t **random** component of $\mathbf{y}_t$

Components

- ★ Long-term: the **trend**
- ★ Periodic: the **seasonality**
- ★ Random: the **residual**



Components

- ★ Additive scheme $y_t = f_t + s_t + \epsilon_t$
- ★ Multiplicative scheme $y_t = f_t \odot s_t \odot (1 + \epsilon_t)$
(and variations)

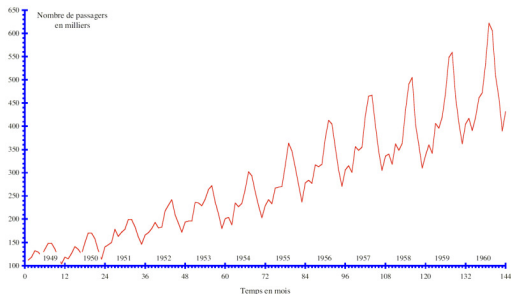


FIG. 4 – Nombre mensuel de passagers internationaux aux États Unis de 1949 à 1960

Note: Multiplicative is equivalent to the additive scheme when considering the log of the series

Component estimation

A rigorous analysis requires the estimation of the trend, independent from seasonality and noise

Smoothing

By definition, s_t and ϵ_t are 0-mean:

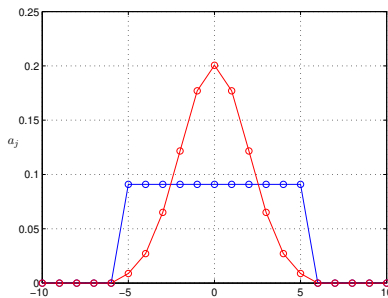
$$\mathbb{E}S = \mathbb{E}E = 0$$

A **moving average** of y_t preserves the trend only

Moving average Ψ_p

$$\Psi_p(\mathbf{y}_z) = \bar{\mathbf{y}}_t = \sum_{j=-p}^p a_j \mathbf{y}_{t-j}$$

with $a_{-j} = a_j$ (symmetric filter) with $\sum_j a_j = 1$.



Note: This is equivalent to a convolution 

Correcting the seasonal variations

- ★ Apply moving average to y_t

$$\Psi_p(y_t) = \Psi_p(f_t) + \Psi_p(s_t) + \Psi_p(\epsilon_t)$$

- ★ if s_t and ϵ_t centered, $\Psi_p(s_t), \Psi_p(\epsilon_t) \approx 0$

$$\hat{f}_t = \Psi_p(y_t)$$

⇒ To “erase” a seasonality with period p , apply moving average Ψ_p

Prediction

- ★ Assume we know $\mathbf{y}_1, \dots, \mathbf{y}_T$ stationary
- ⇒ We wish to infer step $t + p$ ($\mathbf{y}_{t+p}$) from steps until t ($\mathbf{y}_t$) $\rightarrow \hat{\mathbf{y}}_t(p)$
- ★ k is the **horizon**
- ★ In general $\hat{\mathbf{y}}_t(p) \neq \mathbf{y}_{t+p}$ (prediction error) $\rightarrow$ we seek minimum error

No prediction is possible if there is no dependence/causality between values $\mathbf{y}_t$, $t \in \llbracket T \rrbracket$

For the rest of the lecture: $D = 1$ (scalar series):

- ★ Can be generalized per component
- ★ i.e we consider independent components
- ⇒ simplify notation ($y_t \in \mathbb{R}$)

Linear prediction

We assume that y_t depends linearly on the previous values

$$y_t = \sum_{k=1}^n \alpha_k y_{t-k}$$

$$y_t = \boldsymbol{\alpha}^\top \mathbf{y}_{t-1}$$

with $\boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_n]^\top$ and $\mathbf{y}_{t-1} = [y_{t-1}, \dots, y_{t-n}]^\top$

$\Rightarrow$ Linear relationship $\rightarrow$ correlation between values y_t



Stationary process

★ The series is seen as a **stochastic process**

- i.e y_t is a realization (draw/instance) of random variable Y_t
- In general we know **only one** realization of $\{Y_t\}_{t \in [T]}$

⇒ impossible to use correlation between $y_{t-1}, \dots, y_{t-p}$, to predict y_t

Stationary hypothesis (constant mean):

Covariance $\text{cov}(y_t, y_{t-p}) = \gamma_p$ does not depend on t

$$\gamma_p = \text{cov}(y_t, y_{t-p}) = \frac{1}{T-p} \sum_{t=p}^T (y_t - \mu)(y_{t-p} - \mu)$$

Prediction by Exponential smoothing

$$\hat{y}_t(p) = (1 - \tau) \sum_{j=0}^p \tau^j y_{t-1-j}$$

with $\tau \in [0, 1]$

- ★ Most basic method
- ★ The predicted value is the mean of past values
- ★ forget effect (exponential decay) making most recent values important

Exponential smoothing

⌚ Show:

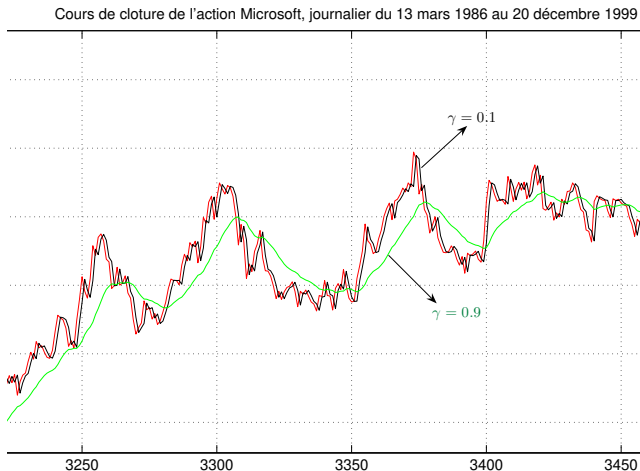
$$\hat{y}_{t+1}(p) = (1 - \tau)y_t + \tau\hat{y}_t(p - 1)$$

or:

$$\hat{y}_{t+1}(p) = \hat{y}_t(p - 1) + (1 - \tau)(y_t - \hat{y}_t(p - 1))$$

- ★ $\tau \rightarrow 1$ prediction accounts for far past (smooth)
- ★ $\tau \rightarrow 0$ prediction only depends on immediate past (less smooth)

Exponential smoothing



Auto-regressive models (AR)

- ★ order- p AR model: $AR(p)$:

$$y_t = \sum_{j=1}^p a_j y_{t-j} + \epsilon_t$$

- ★ Values at t linearly depends the on the p preceding values
- ★ ϵ_t is a white noise with variance σ_ϵ^2 ($\epsilon_t \sim \mathcal{N}(0, \sigma)$)

Parameter estimation

- ★ Mean-squared error regression:

$$\mathbf{a}^* = \underset{\mathbf{a}}{\operatorname{argmin}} \mathbb{E}[e_t^2] = \underset{\mathbf{a}}{\operatorname{argmin}} \sum_{t=0}^T \left(y_t - \sum_{j=1}^p a_j y_{t-j} \right)^2$$

- ★ Solution

$$\frac{\partial \mathbb{E}[e_t^2]}{\partial a_j} = 0 \quad j \in \llbracket p \rrbracket$$

Correlation and regression coefficient

★ We get for j :

$$\begin{aligned}\frac{\partial E[e_t^2]}{\partial a_j} &= 2\mathbb{E}_t \left[\left(y_t - \sum_{i=1}^p a_i y_{t-i} \right) y_{t-j} \right] \\ &= \gamma_j - \sum_{i=1}^p a_i \gamma_{i-j} = 0\end{aligned}$$

★ Recall $\gamma_j = \text{cov}(y_t, y_{t-j}) = \gamma_{-j}$ and y_t stationary

Yule-Walker Equations

Let $\rho_k = \frac{\gamma_k}{\gamma_0}$ be the **correlation function**

$\Rightarrow$ the minimization of mean squared error leads to

$$\rho_k = \sum_{i=1}^p a_i \rho_{k-i} \quad k > 0$$

$$\left\{ \begin{array}{ll} \rho_1 = a_1 \rho_0 + a_2 \rho_1 + a_3 \rho_2 + \cdots + a_p \rho_{p-1} & k = 1 \\ \rho_2 = a_1 \rho_1 + a_2 \rho_0 + a_3 \rho_1 + \cdots + a_p \rho_{p-2} & k = 2 \\ \vdots & \vdots \\ \rho_p = a_1 \rho_{p-1} + a_2 \rho_{p-2} + a_3 \rho_{p-3} \cdots + a_p \rho_0 & k = p \end{array} \right.$$

Auto correlation matrix

In matrix form, Yule-Walker equations are:

$$\mathbf{R}_p \mathbf{a} = \boldsymbol{\rho}$$

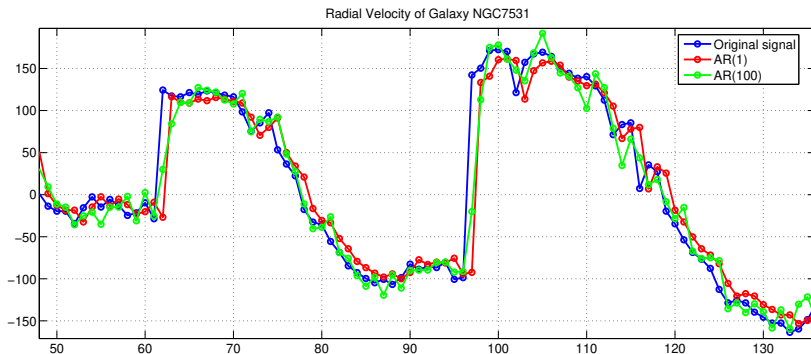
with $\mathbf{a} = [a_1, \dots, a_p]^\top$, $\boldsymbol{\rho} = [\rho_1, \dots, \rho_p]^\top$ and

$$\mathbf{R}_p = \begin{pmatrix} 1 & \rho_1 & \rho_2 & \rho_3 & \dots & \rho_{p-1} \\ \rho_1 & 1 & \rho_1 & \rho_2 & \dots & \rho_{p-2} \\ & \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho_{p-1} & \rho_{p-2} & \rho_{p-3} & \dots & \dots & 1 \end{pmatrix}$$

If $\mathbf{R}_p$ is invertible (positive definite)

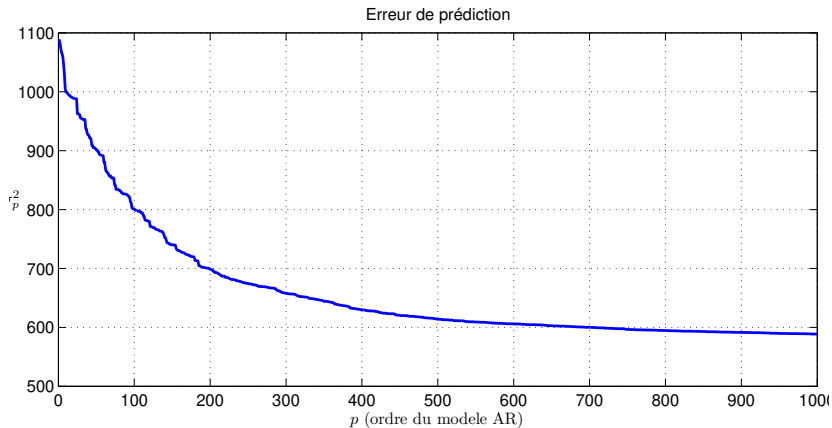
$$\mathbf{a} = \mathbf{R}_p^{-1} \boldsymbol{\rho}$$

Example



⇒ Choice of **optimal** order p ?

Example



⇒ Choice of **optimal** order p ?

Order selection

The higher p the better, but :

- ★ Requires the estimation of p correlation coefficients $\rightarrow$ Confidence over a finite series?
- $\Rightarrow$ Rule of thumb for length N : $p < N/3$
- ★ Model parsimony : in applications, it is generally better to keep a simple model
- $\Rightarrow$ Define parsimony (low p ?)

Order selection

Whitening test (AR model)

$$\epsilon_t = y_t - \sum_{i=1}^p a_i y_{t-i} \quad \text{white noise? (i.i.d. } \sim \mathcal{N}(0, \sigma))$$

⇒ Tests: Durbin-Watson, Fisher,...

★ Check whether values ϵ_t ($t \in \llbracket T \rrbracket$) are **significantly decorrelated**

Final Prediction Error (FPE)

Definition of a global error, combining the variance of the prediction error σ_p and a variance over the imprecision of parameter estimation:

$$\text{FPE}(p) = \sigma_p \cdot \frac{T + p + 1}{T - p - 1}$$

decrease with $p \longleftrightarrow$ increase with p

Criteria for parsimony

Generally from Information Theory :

- ★ Minimum Description Length (MDL)

$$\text{MDL}(p) = N \log(\sigma_p^2) + p \log(T)$$

- ★ Akaike Information Criterion (AIC)

$$\text{AIC}(p) = N \log(\sigma_p^2) + 2(p + 1)$$

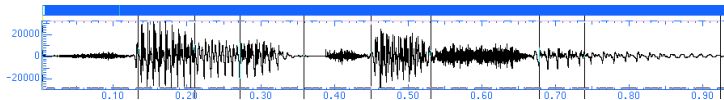
⇒ MDL is a good criterion with low number of samples

Limitations

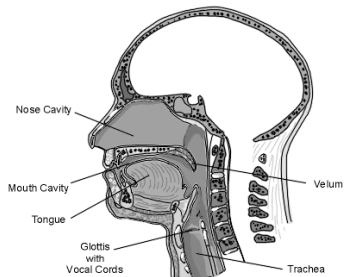
- ★ **Stationarity** is a strong assumption
- ★ Econometric data is rarely stationary (trend, multiplicative seasonality)
- Trend removal, log model, ...
- More complex models (ARMA, ARIMA, ARCH...) at the price of a more complex estimation
- ★ Example of stationary signal: **speech** (at least locally)

Speech analysis and compression using AR models

Hypothesis: local stationarity



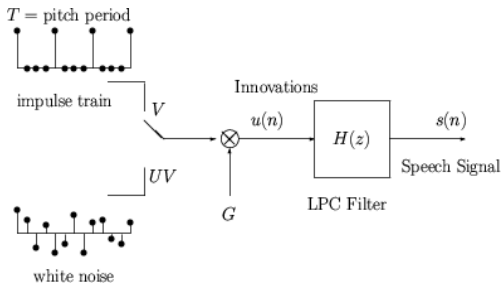
Linear model for speech production



Air flow :

- ★ Excited by vocal chords
- ★ Echo effect on vocal cavity (linear system)

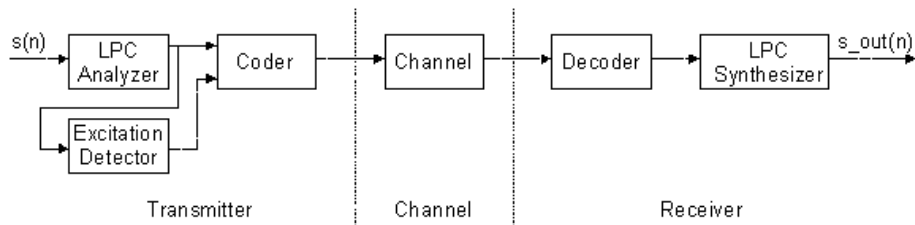
Modeling



- ★ Can show that coefficients $H(z)$ are the parameters of the AR model
- ★ Characterize the parameters of **speech production**
- ★ Important for **speech analysis and compression**

Compression by Linear Predictive Coding (LPC)

Vocoder



⇒ used in GSM

Summary

- ★ As opposed to static data, temporal data includes **causality**
 - ★ Temporal data present a **unique** sample
 - ★ Correlation is found by considering **stationarity**
- ⇒ need to remove trend and seasonality
- ★ $AR(p)$ models are the base for temporal signal modeling
 - ★ They are **linear** (latent) models (coefficients are used for interpretation, generation, ...)
 - ★ Extension to Neural Nets for sequential data: e.g [2] (chap 15)

Example questions [mostly require formal – mathematical – answers]

- ★ What are trend, seasonality and how to remove them? What are the assumptions made?
- ★ What is stationarity? What does it imply and allow?
- ★ What is exponential smoothing (and exponential decay)?
- ★ Provide the base models for linear auto-regression
- ★ What is the horizon (and the lag)?

Ⓢ It is strongly advised to develop the algebra contained in this chapter
(inc relation to auto-correlation and convolution)

References I

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Springer-Verlag, Berlin, Heidelberg, 2006. (available online).
- [2] Kevin P. Murphy. *Probabilistic Machine Learning: an Introduction*. MIT Press, 2022. (available online).
- [3] Robert H. Shumway. *Time Series Analysis and its Applications, with R Examples*. Springer, 2017. (available online).